MarketSight

Stock Market Prediction System

Testing Plan & Acceptance Test

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COSC 471 - Winter 2026 | Okanagan College

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1. Test Strategy Overview

1.1 Testing Levels

Level	Scope	Tools	Responsibility
Unit Tests	Individual functions and methods	Pytest, Pytest-mock	Developers
Integration Tests	API endpoint request/response cycles	Pytest, FastAPI TestClient	Developers
Regression Tests	CI pipeline on every push/PR	GitHub Actions	Automated
Acceptance Tests	End-to-end user workflows	Manual testing	Zane & Dante

1.2 Test Environment

Environment	Configuration
Unit/Integration	In-memory SQLite via pytest fixtures
Regression (CI)	Ubuntu Latest, GitHub-hosted runner
Acceptance	Docker Compose (full stack) on local machine

2. Test Case Specification

2.1 Unit Test Cases

Market CRUD (test_market_crud.py)

TC-ID	Test Case	Input	Expected Output
TC-U01	Create a stock record	Stock(symbol="AAPL", name="Apple")	Stock created in DB
TC-U02	Read a stock by symbol	get_stock(session, "AAPL")	Returns Stock object
TC-U03	Read non-existent stock	get_stock(session, "INVALID")	Returns None
TC-U04	Create daily price record	DailyPrice(open, high, low, close, volume)	Price record created
TC-U05	Read OHLC data for symbol	get_ohlc(session, "AAPL", days=5)	Returns list of 5 prices
TC-U06	List stocks with filtering	list_stocks(session)	Returns all seeded stocks

Market Service (test_market_service.py)

TC-ID	Test Case	Input	Expected Output
TC-U07	List stocks via service	MarketService.list_stocks()	Returns stock list
TC-U08	Get stock via service	MarketService.get_stock("AAPL")	Returns Stock
TC-U09	Get OHLC via service	MarketService.get_ohlc("AAPL", 5)	Returns OHLC list

Confidence Scoring (test_confidence.py)

TC-ID	Test Case	Input	Expected Output
TC-U10	Volatility score - low	volatility=0.01	Score near 1.0
TC-U11	Volatility score - high	volatility=0.10	Score near 0.0
TC-U12	RSI score - neutral (50)	RSI=50	Score near 1.0
TC-U13	RSI score - overbought (80)	RSI=80	Score < 0.5
TC-U14	Volume score - normal	volume_ratio=1.0	Score near 1.0
TC-U15	Return magnitude - small	return=0.01	Score near 1.0
TC-U16	Composite confidence	All factors combined	Score between 0.0-1.0

TC-U17	Edge case - NaN features	NaN values in features	Handles gracefully

2.2 Integration Test Cases

Health Check (test_utils.py)

TC-ID	Test Case	Input	Expected Output
TC-I01	Health check endpoint	GET /api/v1/utils/health-check/	200, body: true

Market API (test_market.py)

TC-ID	Test Case	Input	Expected Output
TC-I02	List stocks endpoint	GET /api/v1/market/stocks	200, returns stock array
TC-I03	Get stock by symbol	GET /api/v1/market/stocks/AAPL	200, returns stock
TC-I04	Get OHLC data	GET /api/v1/market/stocks/AAPL/ohlc	200, returns OHLC array
TC-I05	Invalid stock 404	GET /api/v1/market/stocks/INVALID	404

Inference API (test_inference.py)

TC-ID	Test Case	Input	Expected Output
TC-I06	Predict stock price	GET /api/v1/inference/predict/AAPL	200, PredictionResponse
TC-I07	Invalid stock prediction	GET /api/v1/inference/predict/INVALID	404
TC-I08	Response schema check	Prediction response fields	All required fields present

3. Acceptance Test Specification

Tested by: Zane Tessmer & Dante Bertolutti **Date:** March 15, 2026 **Environment:** Docker Compose (full stack), Chrome browser

3.1 Acceptance Test Cases

AT-ID	Category	Test Description	Expected Result	Actual Result	Status
AT-01	Dashboard	Navigate to /dashboard, verify 4 summary cards	Cards: stocks, sectors, model, API status	Displayed correctly	**PASS**
AT-02	Stocks	Navigate to /dashboard/stocks, search "AAPL"	AAPL appears in filtered list	Found and displayed	**PASS**
AT-03	Chart	Click AAPL, verify candlestick chart	Interactive chart with OHLC data	Rendered with range breaks	**PASS**
AT-04	Chart	Toggle to line chart mode	Line series displayed	Line chart displayed	**PASS**
AT-05	Chart	Select 1W time range	Chart zooms to 7 days	Applied correctly	**PASS**
AT-06	Chart	Enable prediction overlay	Orange dashed line to predicted price	Overlay rendered	**PASS**
AT-07	Predictions	Navigate to /dashboard/predictions	Page with stock selector	Loaded correctly	**PASS**
AT-08	Predictions	Select AAPL, click Predict	Card: symbol, prices, return, confidence	All fields populated	**PASS**
AT-09	Predictions	Verify confidence range	Score between 0-100%	Within valid range	**PASS**
AT-10	Predictions	Click "Predict All"	Cards for all stocks	All predictions shown	**PASS**
AT-11	Predictions	Verify confidence colors	Green >70%, Yellow 40-70%, Red <40%	Colors matched	**PASS**
AT-12	API	GET /predict/INVALID	404 error	"Stock not found"	**PASS**
AT-13	API	GET /utils/health-check/	200, true	Response correct	**PASS**
AT-14	CI	Run CI pipeline on main	All 6 jobs pass	All green	**PASS**

AT-15	Docker	docker compose up	All services healthy	All running	**PASS**
AT-16	Training	Navigate to /dashboard/training	Training monitor loads	Page loaded	**PASS**
AT-17	Streamlit	Open Streamlit app	Market data visible	Data displayed	**PASS**
AT-18	Datetime	Check timestamps across pages	All show UTC	UTC consistent	**PASS**
AT-19	Model	Verify model loads (native .ubj)	Loads within 2s	Loaded in ~1.2s	**PASS**
AT-20	Deployment	Access via production URL	App served with SSL	Served correctly	**PASS**

4. Test Incident Report

Incident #1: Confidence Display Null Check

Field	Details
Incident ID	TI-001
Date	March 15, 2026
Test Case	AT-09 (Verify confidence range)
Severity	Medium
Description	Confidence score was not displaying on prediction cards. The field showed blank instead of a percentage.
Root Cause	The predictions page had a null check `if (confidence !== null)` but the rebased code was returning `confidence` as `undefined` instead of `null` due to a type mismatch between the old and new API response shapes.
Resolution	Updated the null check to `if (confidence != null)` (loose equality) to handle both null and undefined.
Fixed By	Dante Bertolutti
Verified By	Zane Tessmer

Incident #2: Chart Range Breaks on Weekends

Field	Details
Incident ID	TI-002
Date	March 10, 2026
Test Case	AT-03 (Candlestick chart rendering)
Severity	Low
Description	Candlestick charts showed large gaps on weekends where no trading occurred, making the chart hard to read.
Root Cause	The chart x-axis was using continuous time, including non-trading days.
Resolution	Added range breaks to the x-axis configuration to skip weekends and holidays.
Fixed By	Harsh Kumar
Verified By	Dante Bertolutti

5. Test Summary Report

5.1 Test Execution Summary

Test Level	Total Tests	Passed	Failed	Pass Rate
Unit Tests	17	17	0	100%
Integration Tests	8	8	0	100%
Acceptance Tests	20	20	0	100%
Total	**45**	**45**	**0**	**100%**

5.2 Test Incidents Summary

Total Incidents	Resolved	Open
2	2	0

5.3 CI Pipeline Status

All 6 CI jobs passing on main branch:

- backend-lint: **PASS** (~30s)
- backend-typecheck: **PASS** (~45s)
- backend-test: **PASS** (~2 min, 25 tests)
- frontend-lint: **PASS** (~20s)
- frontend-build: **PASS** (~45s)
- docker-build: **PASS** (~3 min)

5.4 Test Conclusion

All 45 test cases across unit, integration, and acceptance testing levels have passed. Two test incidents were identified and resolved during the testing phase. The system meets the acceptance criteria for the IOC milestone.

5.5 Acceptance Sign-Off

All 20 acceptance test cases passed. The system is approved for Initial Operational Capability.

Signed:

Client Representative: _____ Date: _____

Product Owner (Harsh Kumar): _____ Date: _____

Scrum Master (Zane Tessmer): _____ Date: _____

Tester (Dante Bertolutti): _____ Date: _____